

Worksheet 4: Floral Whorls & Placentation

NCERT Class 11 Biology — Chapter 5: Morphology of Flowering Plants

Student Name: _____	Roll No.: _____	Section: _____
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Section A: Multiple Choice Questions (Select the single best answer)

1. In vexillary (papilionaceous) aestivation of Pea and Bean, the correct arrangement from posterior to anterior is: [_____]

- A) 2 Keels overlap 2 Wings, which overlap 1 Standard
- B) 1 Standard (Vexillum) overlaps 2 lateral Wings (Alae), which overlap 2 anterior fused Keels (Carina)
- C) 5 petals overlap continuously in one direction
- D) Valvate sepals alternate with twisted petals without overlap

2. When stamens are united into two distinct bundles, as seen in the (9)+1 arrangement of Pea, the condition is termed: [_____]

- A) Monoadelphous
- B) Diadelphous
- C) Polyadelphous
- D) Syngenesious

3. Stamens adhered to the petals (epipetalous) and to the perianth (epiphyllous) are found respectively in: [_____]

- A) Brinjal and Lily
- B) Lily and Brinjal
- C) Mustard and China rose
- D) Pea and Citrus

4. Free central placentation, where ovules are borne on the central axis and septa are completely absent, is characteristic of: [_____]

- A) Mustard and Argemone
- B) China rose and Tomato
- C) Dianthus and Primrose
- D) Sunflower and Marigold

Section B: Complete the Placentation Classification Table

Placentation Type	Anatomical Description / Chambering	Standard NCERT Examples
Marginal	Placenta forms a ridge along ventral suture of unilocular ovary	_____
Axile	Multilocular ovary; ovules attached to central axis column	_____
Parietal	Ovules on inner peripheral wall; 2-chambered via replum	_____
Free Central	Ovules on central column; septa completely lacking	_____
Basal	Placenta at base of unilocular ovary; single ovule attached	_____

Section C: Fill in the Blanks & Technical Terminology

5. A sterile, non-functional stamen is termed a _____.

6. Variation in the length of stamen filaments within the same flower is observed in _____ and _____.

7. Aestivation in *Calotropis* is _____, whereas in *Cassia* and *Gulmohur* it is _____.

Section D: Short Answer & Analytical Questions

8. Explain the four modes of stamen cohesion (Polyandrous, Monodelphous, Diadelphous, Polyadelphous) with one plant example for each.

9. Contrast Apocarpous Gynoecium vs. Syncarpous Gynoecium with representative examples.

Lecturer Signature: _____